| Question | | | Marking details | Marks available | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | AO1 | AO2 | AO3 | Total | Maths | Prac |
| 4 | *(a)* | (i) | Force is upward <u>on the magnet</u> (1)<br>[F]LHR quoted **and** application of N3 (1)<br>Force is down on the wire (1) | 1 | 1<br><br>1 | | 3 | | 3 |
| | | (ii) | Mean calculated = 0.0236 or 0.02355 (1) accept $2.4 \times 10^{-2}$<br>Absolute uncertainty calculated or implied (± 0.0012) (1)<br>% uncertainty = 5% (1) accept 1 or 2 sig figs<br>accept range 4 – 5.1% | | 3 | | 3 | 3 | 3 |
| | | (iii) | Gradient = $Bl$ used or implied (1)<br>$B$ = 0.295 [T] (no sig fig penalty) (1)<br>Adding % uncertainties **ecf from (a)(ii)** (11%) (no sig fig penalty) (1) accept first principles approach<br>(0.295 ± 0.032)T or (0.30 ± 0.03) **ecf** must have consistency of sig figs either 1 or 2 sig figs in uncertainty + **UNIT** or equivalent in mT (1) | | 4 | | 4 | 4 | 4 |
| | *(b)* | (i) | Force on charge carriers due to mag field or [F]LHR (1)<br>Charge carriers go to one side of conductor / reference to build-up of charge (1)<br>Electric field set up or top is +ve / -ve depending on carriers (1) | 3 | | | 3 | | 3 |
| | | (ii) | **Indicative content:**<br>1. Line goes through origin.<br>2. Graph is a straight line.<br>3. Line passes through all the error bars.<br>4. Both values of $B$ compared.<br>5. Hall probe is more precise or its uncertainty is smaller [accept accurate].<br>6. Sensible comment comparing methods e.g. method A is graphical, method B required only one measurement or converse, method A is more prone to human error, method B doesn't confirm $F = BIl$ | | | 6 | 6 | 3 | 6 |